

KUNAL KARANKAL

CAD Automation Engineer — Mechanical Design Engineer

Hinjewadi, Pune, Maharashtra

☎ +91 9834552744 ✉ mr.karankalkunal@gmail.com 🔗 [linkedin.com/in/karankalkunal](https://www.linkedin.com/in/karankalkunal) 🌐 kcv.vercel.app

Professional Summary

Mechanical Design Engineer with nearly 3 years of full-time experience, including 10 months in CAD automation. Skilled in Autodesk Inventor, VB.NET/iLogic configurators, parametric design, 3D assemblies, manufacturing drawings, BOM generation, GD&T, and sheet metal.

Experience

Pall Corporation

Nov. 2025 – Present

Design Engineer

Pune, Maharashtra

- Supported and developed configurator-based automation using **VB.NET** and **iLogic** to generate ready-to-use **Autodesk Inventor** CAD models and drawings from input parameters, reducing manual effort and design cycle time.
- Standardized engineering inputs, CAD rules, and housing configuration logic to improve repeatability and reduce manual design work.
- Collaborated with R&D, commercial, quality, and production teams on product configuration, cycle-time reduction, **VAVE**, and process improvement.

CG Power and Industrial Solutions Ltd

Dec. 2023 – Nov. 2025

R&D Design Engineer

Nashik, Maharashtra

- Supported nine **36 kV RMU** variants and 12/36 kV switchgear products using **Autodesk Inventor** and **AutoCAD**, producing 3D assemblies and 2D manufacturing drawings.
- Designed RMU cladding, copper connections, mechanisms, safety interlocks, and fabrication-ready **sheet metal** parts.
- Developed adaptation kits for **AIS retrofits**; applied **GD&T**, tolerance analysis, and BOM generation for manufacturing release.
- Coordinated prototypes, fit-form-function checks, shop-floor feedback, benchmarking, and design improvements for manufacturability and cost efficiency.

CG Power and Industrial Solutions Ltd

Sep. 2021 – Oct. 2021

Intern

Nashik, Maharashtra

- Performed 3D modeling and component analysis in **Creo** and studied materials and finishing methods for switchgear applications.

Technical Skills

CAD Automation: VB.NET, iLogic, Autodesk Inventor Automation, Parametric Modeling, Product Configurators, BOM Generation

CAD / Tools: Autodesk Inventor, AutoCAD, Creo, SolidWorks

Mechanical Design: 3D Modeling, 2D Manufacturing Drawings, Product Assemblies, Sheet Metal, GD&T, Tolerance Analysis, DFM, Prototyping, VAVE, Design Validation

Analysis / Languages: Ansys Fluent, HyperWorks, MATLAB (Basic), CFD; English, Hindi, Marathi, Japanese (Basic)

Education

SVKM's Institute of Technology, Dhule

Sep. 2020 – May 2023

B.Tech in Mechanical Engineering, CGPA: 8.72

Dhule, Maharashtra

R. C. Patel Polytechnic, Shirpur

Sep. 2017 – May 2020

Diploma in Mechanical Engineering, 83.90%

Shirpur, Maharashtra

Projects

Mechanical Design & CAD Automation Portfolio

2026

- Built a portfolio showcasing selected CAD automation and mechanical design work.

Ranker Protocol – Android Productivity App

2026

- Developed a private, offline app for tasks, habits, health, finances, goals, reminders, and protected local records, with no advertising, analytics, or cloud synchronization.

Design, Fabrication, and CFD Analysis of Fin Configurations

May 2023

- Improved heat transfer coefficient by **15%** through fin geometry optimization using CFD analysis and experimental validation.

IoT-Based Energy Management System — Smart India Hackathon

Jun. 2022

- Developed an Android application using **Arduino**, **Android Studio**, and **PyCharm**, reducing energy evaluation time by **30%** with real-time graphical reporting.

Publication

Co-author, Scopus-indexed Springer chapter: *Computational Analysis of Various Fin Configurations—A Comprehensive Assessment*, 2024, pp. 117–127.